Yuankai (William) He

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Research Agenda

Physical intelligence for autonomous mobility: latency-aware AV runtimes, programmable roadsides, and digital twins. My research builds the **systems layer** for embodied AI with **measurable timing and safety evidence**.

Education

- **University of Delaware**, Ph.D. in Computer Science. Advisor: Weisong Shi. Sept. 2021 – Current
- **Wayne State University**, M.S. in Robotics. Advisor: Weisong Shi. Sept. 2020 – May 2023
- **Boston University**, B.S. in Computer Engineering. Sept. 2016 – May 2019

Faculty-Candidate Highlights

- **Systems Platform Builder:** Built and deployed the **autonomy software stack** for **two UD AV robotaxi platforms**; built AV/CPS platforms across middleware, infrastructure, system modeling, OS control, and digital-twin testbeds.
- **Patent & Cited Journal Impact: Issued patent** in cyber-physical testing; first-author *IEEE T-VT* article on C-V2X collaborative autonomy has **70 Google Scholar citations** since publication in 2023.
- **Book-to-Curriculum Leadership:** Co-authored a *Springer autonomous-driving textbook* and created a graduate course now **required** in UD's M.S. in AI Autonomous Driving.
- **Top Departmental Recognition & Visibility: Frank A. Pehrson Award**, UD CIS's top graduate student honor (2026); talks at *IEEE IV, UCI CADRI, William & Mary, and AutoSens Detroit*.
- **External Funding & Industry Traction: Lead graduate researcher and proposal contributor** on about **$871,000** in externally supported academic and industry research with Tier IV, Toyota, and NSF.
- **Open-Source Contributions: Autoware Foundation API Workgroup Lead**; TSC member shaping middleware, V2X, and autonomy roadmap decisions.

Research Systems and Platforms

- **SIM runtime**. Built a *freshness-aware ROS 2/Autoware middleware layer* deployed on **two UD AVs** for robotaxi research; up to **98% lower transport latency**, **96% lower p99 latency**, and **13.6 ft shorter braking distance** at 40 mph.
- **OpenIntersection / IPI roadside platform**. Built a *software-defined roadside stack* and API deployed with **two UD AVs** for robotaxi research; links V2X, roadside perception, and planning with **sub-30 ms advisory latency**.
- **DaVoS AV system model**. Built an *instantaneous AV system model* that captures conflict dynamics, system latency, and uncertainty; provides **per-scenario safety evidence** and makes failures traceable across perception, planning, and control.
- **DAVOS operating system**. Built the *Delaware AV Operating System* for **two UD AVs** used in robotaxi research; user-level kernel monitors AV tasks and modules, sets priorities, scheduling policies, and CPU affinity, and supports module swaps.
- **BlueICE / ICAT testbeds**. Built *digital-twin and co-simulation infrastructure* for CARLA, SUMO, AV stacks, and hardware-in-the-loop evaluation; bridges simulation, AV software, and physical testbeds for repeatable validation.

Publication Profile

- Google Scholar (as of May 2026): **118 citations**.

Awards & Honors

- **Frank A. Pehrson Graduate Student Award for Outstanding Computer Science Research** (2026), University of Delaware. Top graduate student honor in UD CIS.
- **Daniel L. Chester Graduate Fellowship Award** (2025, 2026), University of Delaware. Annual CIS fellowship; consecutive-year recipient.

Publications

Books

- W. Shi, **Y. He**. *Introduction to Autonomous Driving*. Springer, 2025. ISBN 3031994841.

Journal Articles

- **Y. He**, W. Shi. "A Vision for Transformative Intersections." *IEEE Computer*, 57(12):58–68, 2024. `doi:10.1109/MC.2024.3421963`
- **Y. He**, B. Wu, Z. Dong, J. Wan, J. Zhang, W. Shi. "Towards C-V2X Enabled Collaborative Autonomous Driving." *IEEE Transactions on Vehicular Technology*, 72(12):15450–15462, 2023. Google Scholar: 70 citations since publication. `doi:10.1109/TVT.2023.3299844`

Conference and Workshop Publications

- **Y. He**, W. Shi. "A Faster and More Reliable Middleware for Autonomous Driving Systems." To appear in *IEEE Intelligent Vehicles Symposium (IV)*, 2026.
- **Y. He**, H. Chen, I. Safro, W. Shi. "A Graph-Based Metric on Dataset Diversity and Redundancy Evaluation." To appear in *IEEE Intelligent Vehicles Symposium (IV)*, 2026.
- **Y. He**, Y. Wang, B. Tian, W. Shi. "DaVoS: An Instantaneous Safety Model for Autonomous Vehicles." To appear in *Physical Intelligence: Systems and Applications (PISA) Workshop*, 2026.
- Y. Wang, **Y. He**, R. Wang, W. Shi. "Quantitative Analysis of Storage Requirement for Autonomous Vehicles." *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '24)*, pp. 71–78, 2024. `doi:10.1145/3655038.3665948`
- Z. Tian, **Y. He**, B. Tian, R. Zhong, E. Foorginejad, W. Shi. "ICAT: An Indoor Connected and Autonomous Testbed for Vehicle Computing." *IEEE International Conference on Mobility, Operations, Services and Technologies (MOST)*, pp. 242–250, 2024. `doi:10.1109/MOST60774.2024.00033`
- B. Wu*, **Y. He***, Z. Dong, J. Wan, J. Zhang, W. Shi. "To Turn or Not To Turn, SafeCross is the Answer." *IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)*, pp. 414–424, 2022. (*Equal contribution) `doi:10.1109/ICDCS54860.2022.00047`

Preprints

- **Y. He**, H. Chen, W. Shi. "An Advanced Framework for Ultra-Realistic Simulation and Digital Twinning for Autonomous Vehicles." *arXiv:2405.01328*, presented at TRB Annual Meeting 2025.
- **Y. He**, W. Shi. "CARScenes: Semantic VLM Dataset for Safe Autonomous Driving." *arXiv:2511.10701*.
- Y. Wang, **Y. He**, W. Shi. "AVS: A Computational and Hierarchical Storage System for Autonomous Vehicles." *arXiv:2511.19453*.
- Y. Wang, **Y. He**, B. Tian, L. Xian, W. Shi. "DAVOS: An Autonomous Vehicle Operating System in the Vehicle Computing Era." *arXiv:2601.05072*.

Courses Taught

- **Introduction to Autonomous Driving**, University of Delaware, CIS.
  Graduate course creator and instructor; now required in the M.S. in AI (Autonomous Driving concentration). Inaugural offering enrolled 20+ students from multiple departments. Fall 2025
- **Introduction to Mobile Robot Programming**, University of Delaware, Department of Computer and Information Sciences.
  Instructor. Summer 2024
- **Autonomous Driving Academy**, University of Delaware.
  Course developer and instructor. Summer 2025

Mentoring

- **CISC Graduate Promotion Unit (GPU)**, University of Delaware.
  Founder and President; built graduate and alumni mentoring pipelines. 2024 – present
- **VIP Mentor**, University of Delaware.
  Mentored Dustin (Duy Duc) Tran. 2024
- **DSU–UD Summer Engineering Research Experience**.
  Mentored Jared Bryant. 2024

Proposal Writing & Funding

- **Funding engagement**. Approximately $180,000 in industry-supported work (Tier IV, Toyota) and approximately $691,000 in government/NSF-supported work.
  Lead graduate researcher and proposal contributor.
- **Tier IV**. *Design and Implementation of Infrastructure Programming Interface (IPI)*.
  Proposal contributor; Lead Graduate Researcher. 11/2024 – 10/2025
- **Tier IV**. *Investigating the Effect of C-V2X on Autoware.Universe*.
  Proposal contributor; Lead Graduate Researcher. 09/2023 – 08/2024
- **Toyota**. *Open Intersection Reference Design and Implementation*.
  Proposal contributor; Lead Graduate Researcher. 01/2024 – 12/2024
- **NSF**. *Deterministic Communication and Predictive Computation for Connected Autonomous Driving as a Service*.
  Proposal contributor; Lead Graduate Researcher. 03/2025 – 02/2028
- **NSF CNS**. *Enabling Real-time, Scalable and Secure Collaborative Intelligence on the Edge*.
  Proposal contributor. 01/2022 – 12/2024

Invited Talks

- "Physics-Aware System Simulation for Intelligent Vehicles."
  Digital Twins for Intelligent Vehicles: From Agent-Level to System-Level Twins,
  IEEE Intelligent Vehicles Symposium. 2026
- "Digital Twins for Autonomy Verification and Validation."
  UCI CADRI Workshop, University of California, Irvine. 2025
- "The Need for Digital Twins and Simulation for Autonomous Vehicle Testing."
  College of William & Mary. 2024
- "Autoware for Autonomous Vehicle Development."
  AutoSens Detroit, Detroit Auto Show. 2023, 2024

Community Leadership & Engagement

- **Autoware Foundation**. API Workgroup Lead; Technical Steering Committee Member; Strategic Planning Committee Member, contributing to middleware and V2X roadmaps in a major open-source autonomy ecosystem.

- **IEEE MOST 2025**. Field Demo Chair.
- **SE4ADS at ICSE 2025**. Program Committee Member.
- **Reviewer**. AAAI 2025/2026; CVPR 2024/2025; TRB Annual Meeting 2025/2026; IEEE WCNC 2025; IEEE Vehicular Technology Conference 2024/2025; IEEE SEC 2024; IEEE MOST 2024; *ACM Computing Surveys* 2024/2025; *Multimedia Tools and Applications* 2022–2025.

Systems and Infrastructure

- **DaVoS**. Developed an instantaneous, platform-agnostic AV system model for matched scenario evaluation. The model compares conservative distance-to-conflict with distance committed during end-to-end latency, making low-confidence behavior traceable to prediction/planning, motion policy, timing overhead, or conservatism.
- **DAVOS**. Developed the Delaware Autonomous Vehicle Operating System, a user-level kernel for AV pipelines deployed on two UD autonomous vehicles used as robotaxi research platforms. DAVOS monitors system and module performance, specifies each task's priority, scheduling policy, and CPU set, and supports modular replacement of perception, planning, control, and storage components.
- **SIM**. Designed a freshness-aware shared-memory runtime for ROS 2 and Autoware that replaces copy/serialize messaging with zero-copy exchange, providing a substrate for timing-aware robotics and CPS pipelines deployed on two UD autonomous vehicles used as robotaxi research platforms; achieved up to 98% lower transport latency, approximately 96% lower p99 tail latency, and a 13.6 ft shorter emergency-braking distance at 40 mph on a production-ready Autoware.Universe vehicle.
- **OpenIntersection / IPI**. Led the design of a software-defined roadside stack and programmable interface for distributed CPS, deployed with two UD autonomous vehicles used as robotaxi research platforms; includes auditable V2X-facing services at intelligent intersections and the C-V2X collaborative-autonomy system behind the first-author IEEE T-VT article with 70 Google Scholar citations.
- **BlueICE / ICAT**. Built a containerized digital-twin and co-simulation framework that synchronizes CARLA, SUMO, and Autoware with hardware-in-the-loop support, plus a V2X-enabled testbed for reproducible robotics and full-stack CPS evaluation.

Intellectual Property

- **Patent**. "Method and system for determining test system of mechanical electronic equipment." Patent No. CN 2018114055597; issued Feb. 19, 2019.

Prior Industry Experience

- **Torc Robotics**, Full-Time Co-op ML Engineer L2. Summer 2023
  Data curation pipelines for training AV perception models.
- **Joyson Safety Systems**, Integration Engineer. Oct. 2019 – Oct. 2020
  GM Hands-On-Wheels project; integration and validation for driver-monitoring and steering-wheel interaction systems.
- **UISEE Technology**, R&D Intern, Autonomous Driving Division. Summer 2018
  Autonomous-shuttle cyber-physical testing.